

Evaluating LLMs for Bias in Non-English Languages

A Persian Perspective

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Abstract

Bias in Large Language Models (LLMs) presents a critical challenge for equitable AI deployment, particularly in under-resourced languages such as Persian. Although multilingual models like LLaMA claim cross-lingual generalizability, their internal representations often reflect dominant language priors, leading to misalignment and amplified bias in non-English contexts.

This thesis investigates two core research questions: (1) Do PersianLLaMA and LLaMA differ in how they amplify or mitigate gender, religion, race, and occupation biases in Persian? (2) How do tokenization strategies and language alignment influence the measurement and manifestation of bias in Persian LLMs?

To address these questions, the *StereoSet* benchmark was adopted into a culturally grounded Persian variant and construct a custom prompt-response dataset reflecting Persian linguistic and social structures. Two contrasting models were compared: LLaMA (3.1–8B), a general-purpose multilingual LLM, and PersianLLaMA (13B), a Persian-specific model trained on native corpora.

Bias is quantitatively assessed using Pseudo Log-Likelihood (PLL), Categorical Bias (CB), and Adjusted Perplexity Index (APX). The results show that PersianLLaMA achieves higher fluency, narrower PLL gaps, and lower bias propagation across all evaluated domains, while also displaying higher PLL variance, indicating greater expressivity and linguistic sensitivity. LLaMA, in contrast, exhibits stronger directional bias in English and degraded consistency in Persian due to tokenization misalignment and limited morphological grounding.

This work provides the first cross-lingual PLL-based bias evaluation for Persian using an adapted *StereoSet* framework, offering a reproducible

methodology for fairness assessment in morphologically rich, low-resource languages. The findings highlight the importance of language-specific pre-training for building culturally aligned and bias-aware LLMs.

Abbreviations

Abbreviation	Description
AI	Artificial Intelligence
NLP	Natural Language Processing
ML	Machine Learning
IoT	Internet of Things
GPU	Graphics Processing Unit
CPU	Central Processing Unit
API	Application Programming Interface
LLM	Large Language Model
PLL	Pseudo Log-Likelihood
CB	Contextual Bias
APX	Adjusted Perplexity Index
PPX	Perplexity
BPE	Byte-Pair Encoding
XAI	Explainable Artificial Intelligence
SLR	Systematic Literature Review
PLM	Pretrained Language Model
DEI	Diversity, Equity, and Inclusion
TTS	Text-to-Speech
ASR	Automatic Speech Recognition
mBERT	Multilingual BERT
XML-R	Cross-lingual Language Model – RoBERTa

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1

Introduction

The rapid advancement of Large Language Models (LLMs) has revolutionized natural language processing (NLP), enabling unprecedented capabilities in text generation, translation, and conversational AI (Devlin, 2018). These models leverage vast amounts of text data and complex neural architectures to achieve state-of-the-art performance in various NLP tasks. However, as LLMs are predominantly trained on internet-scale corpora, they inevitably inherit and propagate the biases present in these datasets (Bender et al., 2021). This raises significant concerns regarding fairness, equity, and ethical deployment, particularly in high-stakes domains such as law, finance, and healthcare (Caliskan et al., 2017). Bias in LLMs manifests in multiple forms, including gender bias, racial and ethnic bias, socio-economic bias, and linguistic bias (Nangia et al., 2020). While extensive research has been conducted on bias in English-centric models, the same level of scrutiny has not been applied to lower-resource languages such as Persian (Barkhordar et al., 2024).

Persian, spoken by over 110 million people across Iran, Afghanistan, and Tajikistan, presents unique linguistic and cultural characteristics that influence bias representation in LLMs (Nazarizadeh et al., 2022). Unlike English, Persian is an inflected language with a relatively free word order, a lack of gendered pronouns, and script-based complexities that impact tokenization and embedding representations. These linguistic traits introduce challenges

in NLP model training, often leading to underrepresentation and misalignment in pre-trained multilingual LLMs. The underrepresentation of Persian in large-scale corpora, combined with cultural and historical narratives embedded in available datasets, results in amplified biases when compared to high-resource languages such as English (Saffari et al., 2024).

The reliance on multilingual models such as mBERT and XLM-R, which claim to provide equitable performance across languages, has proven insufficient in mitigating bias in Persian NLP applications. Recent studies indicate that these models exhibit substantial disparities in fairness and accuracy when applied to Persian text, particularly in sentiment analysis, named entity recognition, and question-answering tasks (Pires et al., 2019; Rajabi & Valavi, 2021). This phenomenon arises from the imbalanced pretraining data, where English and other widely spoken languages dominate, leading to skewed tokenization strategies and representation learning mechanisms that do not adequately capture Persian linguistic nuances. Consequently, Persian LLMs often reinforce stereotypes and produce biased or culturally misaligned outputs, which can have severe implications for applications in media, governance, and social discourse.

As Persian NLP applications grow in importance, ensuring that these models do not disproportionately propagate harmful stereotypes is crucial. Bias in LLMs is not merely a technical limitation but a broader ethical concern that impacts digital inclusivity and equitable AI deployment. The presence of structural and morphological complexities in Persian further exacerbates this issue, as standard bias mitigation strategies developed for English or other high-resource languages may not generalize effectively to Persian. Without dedicated frameworks to evaluate and correct these biases, Persian NLP applications risk reinforcing societal disparities and inaccuracies that could have broad socio-political consequences.

In light of these challenges, this study aims to conduct a systematic analysis of bias in Persian LLMs, evaluating their fairness and identifying the most effective bias mitigation strategies. By leveraging benchmark-driven evaluations and comparative analyses with multilingual models, this research seeks to address the critical gaps in Persian NLP fairness, guiding the development of more ethical and unbiased AI-driven language technologies.

1.1 Problem Statement

The increasing reliance on a specific Persian Large Language Model (LLM) for NLP applications underscores critical yet underexplored challenges in bias evaluation and mitigation. While multilingual LLMs often claim cross-linguistic fairness, this Persian model, trained on limited and unevenly distributed corpora, risks amplifying subtle socio-political and cultural biases unique to Persian-speaking contexts (Abbasi et al., 2023; Saffari et al., 2024). Prior studies demonstrate that low-resource settings magnify bias due to uneven data representation and lack of culturally aligned benchmarks (Nadeem et al., 2021). Unlike English-focused models with established fairness frameworks, Persian LLMs lack dedicated tools to detect and address biases arising from Persia’s distinct discourse patterns, such as implicit power dynamics in formal texts, coreference complexity, or regional dialect variations (Abbasi et al., 2023; Mohammadi et al., 2024). Current mitigation techniques, largely designed for high-resource languages, prove insufficient when applied to this model, failing to adapt to its training data’s historical and contextual skews (Mozafari et al., 2024; Ravfogel et al., 2020). This inadequacy threatens equitable deployment in sensitive areas like legal document analysis, sentiment monitoring, and cross-cultural AI systems (Siddique et al., 2024).

This study targets these gaps by analyzing bias within a specific Persian LLM, developing customized mitigation strategies, and constructing a benchmark for fairness evaluation grounded in established resources. While several recent datasets such as PSN (Saffari et al., 2024) and ParsiNLU (Khashabi et al., 2020) provide valuable perspectives on Persian socio-cultural norms and linguistic challenges, the primary evaluation in this work relies on StereoSet (Nadeem et al., 2021), a widely adopted benchmark for measuring stereotypical bias in pretrained language models. By situating StereoSet within a comparative framework and acknowledging complementary resources, this study aims to enhance reliability and reduce cultural misrepresentation in Persian NLP applications (Kaneko et al., 2023; Siddique et al., 2024).

1.2 Research Question

This study examines bias in Persian large language models through a comparative lens, focusing on PersianLLaMA (Abbasi et al., 2023) and LLaMA. The analysis is structured around two complementary research questions. These questions address different but interconnected aspects of bias: the

first concerns the *extent* and *direction* of bias exhibited by the models, while the second concerns the *mechanisms* that shape how such bias is measured and manifested. Thus, the second question is not a subquestion of the first, but an independent inquiry that enables a deeper technical understanding of the patterns observed in RQ1.

RQ1: “Do PersianLLaMA and LLaMA differ in amplifying or mitigating gender, racial, religious, and occupational biases in Persian text processing?”

PersianLLaMA, designed for Persian contexts, may reflect unique socio-cultural biases embedded in its training data, distinct from LLaMA’s broader multilingual foundation (Abbasi et al., 2023; Saffari et al., 2024). Prior evaluations such as StereoSet (Nadeem et al., 2021) and cross-lingual bias studies (Kaneko et al., 2023) show that bias behavior differs across languages and domains, motivating a Persian-focused comparative analysis. Furthermore, fairness mechanisms developed for English frequently underperform in morphologically rich or under-resourced languages like Persian (Mohammadi et al., 2024; Vakili, 2020).

RQ2: “How do tokenization strategies and language alignment influence the measurement and manifestation of bias in Persian LLMs?”

While RQ1 evaluates observable bias differences, RQ2 investigates the structural factors, such as training data composition, tokenization methods, and alignment strategies, that may explain why these differences arise. This distinction is essential because bias scores can only be meaningfully interpreted when the underlying linguistic processing mechanisms are understood. By jointly analyzing LLaMA-3.1-8B (LLaMA) and PersianLLaMA-13B, the study connects empirical bias outcomes with model-internal design choices, addressing gaps highlighted in recent surveys on multilingual fairness and bias propagation in LLMs (Kaneko et al., 2023; Nadeem et al., 2021; Siddique et al., 2024).

1.3 Delimitations

This study confines its scope to evaluating representational bias in two large language models: PersianLLaMA, a Persian-specific LLM, and LLaMA, a general-purpose multilingual baseline. Other models frequently used in Persian NLP research—such as ParsBERT or XLM-RoBERTa—are excluded from

the primary analysis despite their documented performance benchmarks in monolingual settings (Abaskohi et al., 2024). The objective is to isolate and compare the bias propagation characteristics of models trained with divergent linguistic priorities (language-specific versus multilingual) within the Persian context.

The analysis focuses exclusively on textual bias, deliberately omitting modalities such as visual, auditory, or multimodal systems (Barkhordar et al., 2024). Furthermore, only algorithmic and linguistic sources of bias are examined; social, cultural, and institutional dimensions of fairness, such as annotator bias, regulatory frameworks, or user perception, fall outside the scope of this investigation.

Bias mitigation techniques explored in this work are computational in nature, limited to techniques such as data augmentation, prompt engineering, and model fine-tuning. Broader mitigation strategies involving community participation, institutional auditing, or normative policymaking are considered beyond the methodological boundaries of this thesis.

To overcome the limitations of existing English-centric resources like StereoSet, the study extends its contribution by adapting culturally grounded templates and constructing new Persian-specific bias benchmarks based on the StereoSet dataset, including targeted probes for gender-neutral morphology and occupational stereotypes. These benchmarks enable evaluation metrics such as stereotype intensity, contextual bias scores, and adjusted perplexity (APX), which are tailored to assess fairness in low-resource linguistic settings. By concentrating on Persian LLMs and developing targeted bias evaluation resources, this research distinguishes itself from broader multilingual fairness studies.

1.4 Thesis Structure

This thesis is organized into six chapters. Chapter 1 introduces the research problem, research questions, delimitations, and the overall structure of the thesis. Chapter 2 provides a comprehensive review of the existing literature on NLP, LLMs, bias evaluation techniques, dataset curation methodologies, and cultural biases in AI, with a specific focus on Persian language models. Chapter 3 outlines the methodology, detailing the translation and adaptation of an existing bias evaluation dataset, the selection of evaluation metrics, model selection criteria, and the experimental setup. Chapter 4 presents the empirical results, analyzing the performance of models when

tested on gendered prompts and evaluating bias propagation in Persian LLMs. Chapter 5 interprets the results by comparing different models, discussing the broader implications of the findings, and identifying existing limitations. Finally, Chapter 6 concludes the thesis by summarizing key findings, recommending strategies for mitigating bias in Persian LLMs, and suggesting potential directions for future research.

2

Extended Background

2.1 Description of Large Language Models and Their Mechanisms

Large Language Models (LLMs) such as PersianLLaMA and LLaMA are advanced neural networks built on transformer architectures, designed to process and generate human-like text through multi-layered, attention-driven mechanisms. These models consist of stacked transformer blocks, each integrating multi-head self-attention and feed-forward layers, optimized via residual connections and layer normalization to capture long-range dependencies in sequences (Su et al., 2024). With parameter counts ranging from millions to billions (e.g., Llama-3.1-8B with 8 billion, PersianLLaMA - 13B with 13 billion), LLMs employ a two-phase training process: unsupervised pre-training on vast corpora (e.g., web-scraped data, curated texts) to learn general language representations, followed by supervised fine-tuning or alignment (e.g., via reinforcement learning with human feedback, RLHF) for task-specific performance (Ouyang et al., 2022).

Pre-training involves predicting masked tokens or next words in a sequence, optimizing a cross-entropy loss function over billions of tokens. This process embeds statistical patterns into high-dimensional token embeddings (e.g., 768-4096 dimensions), shaped by subword tokenization methods like Byte-Pair Encoding (BPE) or WordPiece (Rust et al., 2021). For Persian, tokenization poses unique challenges due to its agglutinative morphology and

script continuity, often requiring custom vocabularies to mitigate segmentation errors (Saffari et al., 2024). The self-attention mechanism, defined as:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^\top}{\sqrt{d_k}} \right) V \quad (2.)$$

weights token relationships based on co-occurrence frequency, amplifying dominant patterns, including biases, from the training data into the model’s weights.

During inference, LLMs generate text autoregressively, sampling from a probability distribution over the vocabulary (e.g., top-k or nucleus sampling), influenced by temperature and context window size (typically 2048-4096 tokens). Bias manifests in these outputs as skewed probabilities favoring stereotypical associations, measurable via metrics like perplexity (prediction uncertainty), embedding cosine similarity (semantic bias), and stereotype intensity scores (cultural skew) (Ghosh & Caliskan, 2023). Scaling model size enhances performance but exacerbates bias amplification, as larger parameter spaces overfit to dominant cultural narratives in the data (Su et al., 2024). For Persian LLMs, limited corpus diversity and sub-optimal tokenization further distort representations, necessitating detailed evaluation of these technical underpinnings to assess and mitigate bias effectively.

2.2 Theoretical Framework

The study of bias in Large Language Models (LLMs) is rooted in computational linguistics, sociolinguistics, and fairness-aware machine learning. Bias in NLP systems stems from multiple sources, including imbalanced training data, model architecture, and algorithmic decision-making (Bender et al., 2021). To address these issues, this research builds upon existing frameworks in algorithmic fairness, interpretability, and ethical AI to analyze and mitigate bias in Persian LLMs.

2.2.1 The Computational Perspective: Model and data-included Bias

Bias in LLMs emerges from the way models are trained, particularly in their reliance on large-scale textual corpora. The statistical learning process behind deep learning-based NLP models encodes historical and societal

biases present in the data. Pre-trained models such as BERT, GPT, and XLM-R process billions of words from heterogeneous sources, leading to two primary bias sources: data-induced bias and model-induced bias (Caliskan et al., 2017).

Data-induced bias arises from the nature of the training datasets. The inclusion of web-scraped text from sources such as Wikipedia, Common Crawl, and social media ensures a vast data pool but also embeds stereotypical associations and culturally specific biases. Studies have shown that multilingual models such as mBERT and XLM-R fail to distribute training data equally across languages, leading to disproportionate generalization errors in Persian NLP tasks (Kamali et al., 2022). Tokenization inefficiencies exacerbate this problem, as Persian script segmentation differs significantly from English, causing distortions in language representations (Kamali et al., 2022).

Model-induced bias results from architectural and algorithmic choices. The attention mechanisms in transformers emphasize statistically significant token co-occurrences, reinforcing dominant associations even when they are contextually incorrect. This issue is particularly problematic in Persian coreference resolution, where the absence of explicit gender pronouns in Persian leads to English-trained models defaulting to male-gendered interpretations (Mohammadi et al., 2024). These structural biases contribute to the propagation of stereotypes in Persian NLP applications, influencing machine translation, sentiment analysis, and automated decision-making systems.

2.2.2 Algorithmic Fairness and Bias Measurement

Algorithmic fairness is a foundational concept in bias mitigation, emphasizing the need for equitable treatment of diverse demographic groups in machine learning models (Mehrabi et al., 2021). Techniques such as demographic parity, equalized odds, and fairness-aware embeddings provide quantifiable measures of bias in LLMs (Zhou et al., 2019). These methodologies have been studied in high-resource languages, and while some research has explored their applicability to Persian, significant gaps remain in evaluating their effectiveness across diverse linguistic contexts and domain-specific tasks. Metrics such as stereotype association scores, differential representation indices, and fairness-aware perplexity measures offer insights into how biases manifest in Persian text generation tasks. Large-scale bias evaluation

frameworks such as StereoSet (Nadeem et al., 2021) and WEAT (Caliskan et al., 2017) serve as benchmarks for detecting bias propagation in NLP models, yet require custom adaptations for Persian datasets.

2.2.3 Representation Learning and Linguistic Bias

Language models learn representations based on statistical patterns in data, which inherently reflect societal biases (Caliskan et al., 2017). The structure of Persian, with its morphological complexity, script variations, and linguistic ambiguities, introduces additional challenges in ensuring fair representation learning. Studies have shown that multilingual models disproportionately favor high-resource languages, leading to suboptimal performance and increased bias in Persian NLP tasks (Abbasi et al., 2023). Tokenization inefficiencies, limited pretraining corpora, and cultural-specific discourse patterns further exacerbate these issues, necessitating a more targeted approach to bias evaluation and mitigation.

2.3 Characteristics of Biases in the Persian Context

Bias in Persian language models is influenced by several linguistic, cultural, and data-related factors that make it distinct from biases observed in high-resource languages such as English. The Persian language’s unique grammatical structure, sociolinguistic norms, and political landscape contribute to the challenges of mitigating bias in LLMs trained on Persian corpora.

2.3.1 Linguistic Challenges and Bias in Persian NLP

Persian is an inflected language with a rich morphological system that differs significantly from English and other widely studied languages. Unlike English, Persian does not explicitly encode gender in pronouns, which can lead to coreference resolution errors when gender must be inferred from context (Mohammadi et al., 2024). While this grammatical feature does not inherently favor any gender, biased training data often leads LLMs to associate ambiguous references with male-gendered roles, particularly in professional and social contexts, thereby reinforcing gender stereotypes in generated text.

Additionally, Persian exhibits free word order, allowing subject, verb, and object to appear in multiple configurations within a sentence. This flexibility poses challenges for dependency parsing and word embeddings, as model

architectures pre-trained on rigid word order structures (such as those found in English) fail to generalize well to Persian syntax. This misalignment can lead to misinterpretations of contextual dependencies, amplifying biases in sentiment analysis, machine translation, and text classification tasks (Khashabi et al., 2020).

Furthermore, tokenization presents a significant challenge in Persian NLP due to the language’s complex orthographic rules. Persian script does not use spaces between words in the same way as English, resulting in difficulties for LLMs in segmenting words correctly (Kamali et al., 2022). Tokenization errors can introduce bias by disproportionately affecting words related to underrepresented or marginalized groups, reducing their visibility in training corpora. This issue is further compounded when models trained on English-centric tokenization strategies are adapted to Persian without appropriate modifications.

2.3.2 Sociocultural and Ideological Bias in Persian Corpora

The textual corpora used for training Persian LLMs are heavily influenced by the sociocultural and ideological context of the sources from which they are derived. Many Persian datasets are collected from government-affiliated media, religious texts, and state-controlled digital platforms, which inherently embed specific viewpoints and narratives. This leads to political, religious, and ideological biases that are reflected in the language models trained on these sources (Saffari et al., 2024).

Unlike English datasets, which often incorporate diverse perspectives from multiple sources, Persian corpora tend to be more homogeneous and state-influenced, leading to limited representation of minority viewpoints. This imbalance can manifest in NLP applications as biased content moderation, politically skewed text summarization, and restricted sentiment classification models that disproportionately flag dissenting opinions as negative or harmful (Rajabi & Valavi, 2021)

2.3.3 Gender and Occupational Stereotypes

Persian LLMs exhibit pronounced gender biases due to historical and cultural influences embedded in the training data. Studies on stereotypical bias measurement have shown that Persian LLMs tend to associate men with leadership, science, and decision-making roles, while women are more

frequently linked to caregiving, education, and domestic responsibilities (Saffari et al., 2024).

Gender biases in Persian LLMs are further reinforced by limited representation of women in public discourse and media sources included in training datasets. Since gender-neutral terms in Persian default to masculine interpretations, many NLP applications fail to generate gender-balanced translations, recommendations, or responses, leading to disparities in human-computer interactions (Ghosh & Caliskan, 2023).

2.4 Bias in Large Language Models

2.4.1 Studies on Bias in LLMs

Research on bias in LLMs has extensively demonstrated that these models learn and amplify societal stereotypes present in their training data. Studies such as those by Nadeem et al. (2021) in StereoSet and Kurita et al. (2019) in WEAT (Word Embedding Association Test) have provided quantitative methods for measuring bias in models like BERT, GPT, and RoBERTa. These studies reveal that pretrained models exhibit gender, racial, religious, and occupational biases due to the overrepresentation of stereotypical patterns in large-scale corpora.

Earlier work by Bender et al. (2021) in *On the Dangers of Stochastic Parrots* highlights how biases become more pronounced as models scale in size, with larger LLMs disproportionately favoring high-resource languages and dominant cultural narratives. Similarly, Zhou et al. (2019) demonstrates that coreference resolution in NLP systems exhibits strong gender biases, often linking professions such as 'doctor' to male entities while associating 'nurse' with female ones. More recent research by Saffari et al. (2024) extends these insights, showing persistent gender and cultural biases in Persian LLMs due to training data imbalances. These findings underline the necessity for robust bias evaluation and mitigation strategies in LLM development.

2.4.2 Datasets for LLM Bias Evaluation

Benchmarking bias in large language models (LLMs) requires robust evaluation datasets that assess stereotype propagation, fairness in model predictions, and demographic disparities. For English, widely used datasets such as StereoSet (Nadeem et al., 2021) and CrowS-Pairs (Nangia et al., 2020) quantify bias by measuring how models associate stereotypical traits

with demographic groups. Additionally, BBQ (Bias Benchmark for Question Answering) (Parrish et al., 2021) provides a comprehensive framework for evaluating social biases in question-answering systems, offering diverse contexts and demographic dimensions. Another notable resource is WinoBias (Zhao et al., 2018), which focuses on gender bias in coreference resolution tasks, highlighting how models often reinforce gendered stereotypes.

For Persian, bias evaluation remains underdeveloped due to the lack of standardized datasets. However, recent contributions such as the PSN: Persian Social Norms Dataset (Saffari et al., 2024) provide culturally aligned benchmarks for assessing sociopolitical and gender biases in Persian LLMs. Additionally, ParsiNLU (Khashabi et al., 2020) offers a suite of NLP tasks tailored for Persian, though it lacks explicit bias evaluation metrics.

Another critical dataset for bias evaluation in Persian is the Persian Tokenizer Benchmark (Kamali et al., 2022), which examines the impact of tokenization on fairness in NLP tasks. Given that many Persian LLMs rely on suboptimal tokenization strategies, this dataset plays a crucial role in understanding how preprocessing choices contribute to bias. Furthermore, while not specific to Persian, datasets like COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) highlight the importance of fairness in predictive systems, offering lessons for bias mitigation that can be adapted to Persian-language contexts.

To further advance bias evaluation in Persian LLMs, it may be beneficial to draw inspiration from multilingual bias datasets such as XSTREME (Hu et al., 2021) and Bias Benchmark for NLP (Lalor et al., 2022), which provide frameworks for evaluating bias across multiple languages. These resources can serve as a foundation for developing Persian-specific benchmarks that address unique cultural and linguistic nuances.

3

Method

3.1 Research Strategy and Methodology

This study applies a comparative evaluation framework to examine bias in two Persian-capable Large Language Models: PersianLLaMA (13B) (Abbasi et al., 2023), a model aligned with native Persian corpora, and LLaMA-3.1-8B, a general-purpose multilingual model. The analysis focuses on four sensitive domains (gender, religion, race, and occupation) commonly used to expose social bias in LLMs (Milios & BehnamGhader, 2022; Nadeem et al., 2021; Parrish et al., 2022). Controlled prompt-based experiments enable systematic comparison, motivated by findings that multilingual LLMs often struggle to generalize fairness to low-resource and morphologically rich languages such as Persian (Kaneko et al., 2023; Mohammadi et al., 2024; Siddique et al., 2024).

StereoSet (Nadeem et al., 2021) serves as the primary benchmark due to its dual assessment of stereotypical associations and language modeling ability. This is essential because tokenization errors and low fluency can distort bias signals in non-English languages (Kaneko et al., 2023; Milios & BehnamGhader, 2022). Although PSN (Saffari et al., 2024) and ParsiNLU (Khashabi et al., 2020) offer cultural insights, they are not designed for bias evaluation; StereoSet therefore provides a robust and comparable foundation across languages.

To adapt StereoSet for Persian, a Persian-specific template dataset was constructed by (1) retaining the original minimal-pair logic, (2) using PSN’s prompt construction patterns for cultural grounding (Saffari et al., 2024), (3) localizing sentence structure, names, and sociocultural references, and (4) validating all items with native speakers. Bias is measured using pseudo-log-likelihood (PLL), categorical bias (CB), and the Adjusted Perplexity Index (APX), metrics shown to effectively capture contextual and representational bias (Kaneko et al., 2023; Nadeem et al., 2021; Siddique et al., 2024).

A human annotation approach was considered but rejected due to subjectivity, low scalability, and known inconsistencies in annotator agreement (Mohammadi et al., 2024). The synthetic, template-driven method used here provides a reproducible, culturally consistent framework for assessing fairness in Persian LLMs.

Ethical Considerations

This study involves sensitive sociocultural categories such as gender, religion, race, and occupation. Following established guidelines for responsible bias evaluation in NLP (Nadeem et al., 2021; Parrish et al., 2022; Siddique et al., 2024), all prompts and templates were created strictly for diagnostic purposes and do not aim to reinforce or legitimize stereotypes. All Persian examples were synthetically constructed and reviewed by native speakers to ensure cultural appropriateness and to avoid harmful or inappropriate formulations.

No personal or identifiable data was used at any stage, ensuring compliance with standard research ethics. The analysis focuses exclusively on *model behavior*, and all results are reported in aggregate form to prevent misinterpretation or stigmatization of real communities, reflecting concerns highlighted in multilingual fairness research (Milios & BehnamGhader, 2022; Mohammadi et al., 2024).

Potential methodological limitations, such as tokenization artifacts, cultural coverage gaps, and template constraints, are acknowledged in accordance with observations from recent surveys on multilingual and low-resource bias evaluation (Kaneko et al., 2023; Siddique et al., 2024). These measures collectively ensure that the evaluation remains ethically responsible, culturally sensitive, and aligned with best practices in fairness-oriented NLP research.

3.2 Preprocessing Pipeline for Bias Evaluation

To evaluate representational bias in large language models (LLMs) within the Persian linguistic context, a systematic preprocessing pipeline was developed, leveraging the intersentence configuration of the StereoSet benchmark. This framework employs context-dependent sentence completions to quantify implicit biases as reflected in probabilistic model outputs. The pipeline involved curating stereotype-rich samples from the McGill-NLP/StereoSet dataset, focusing on the validation subset with clear bias annotations. These samples were filtered to ensure semantic coherence, aligned with Persian linguistic structures, and reformatted for log-likelihood-based evaluation. High-fidelity Persian translations, manually verified for cultural equivalence and syntactic accuracy, were generated to enable cross-lingual fairness analysis. SentencePiece-based tokenization, tailored to Persian morphology, ensured compatibility with modern transformer tokenizers, facilitating reliable computation of pseudo-likelihood scores, contextual bias variances, and the Adjusted Perplexity Index (APX).

3.3 Dataset Composition and Characteristics

The resulting dataset, termed NewStereoSet (Custom Dataset), comprises 240 contextual prompts, each paired with three semantically distinct completions (stereotype, anti-stereotype, unrelated), yielding 720 sentence-level instances across four bias domains: race, gender, religion, and profession. Each data row includes:

- A contextual sentence (e.g., “The people from Qatar are always BLANK”),
- A target descriptor (e.g., “Qatar”),
- A labeled completion categorized as:
 - Stereotype (label = 0),
 - Anti-stereotype (label = 1),
 - Unrelated (label = 2),
- The full English sentence (context + completion),
- A corresponding Persian translation, ensuring sociolinguistic alignment.

For instance:

- English: “The people from Qatar are always rich” → Stereotype
- Persian: “مردم قطر همیشه ثروتمند هستند”

This bilingual structure supports uniform evaluation of Persian-specialized models (e.g., PersianLLaMA-13B) and multilingual baselines (e.g., LLaMA-3.1-8B) under equivalent semantic conditions, enabling robust cross-lingual bias assessment.

3.4 Dataset Processing Methodology

3.4.1 Data Acquisition and Bias-Focused Filtering

The preprocessing pipeline began with the retrieval of the McGill-NLP/StereoSet dataset via the datasets library, focusing on the intersentence variant with triplet-based completions. Entries were programmatically filtered to retain only those with explicit bias annotations (race, gender, religion, profession), ensuring semantic relevance and alignment with domain-specific bias taxonomies.

Balanced Stratified Sampling

To ensure equitable representation across bias domains, a stratified sampling approach was employed, selecting 60 prompts per category (race, gender, religion, profession). This yielded a balanced dataset of 240 prompts, each mapped to three completions, totaling 720 data points, thus supporting statistically robust comparisons.

3.4.2 Structural Normalization and Semantic Labeling

The hierarchical StereoSet structure was flattened to facilitate annotation, translation, and evaluation. Each completion was extracted as an individual row, assigned a semantic label (stereotype, anti-stereotype, unrelated), and mapped from numerical annotations (0, 1, 2) to human-readable categories for enhanced interpretability. This transformation enabled sentence-level granularity during model inference and scoring.

3.4.3 Cross-Lingual Translation

All completions were translated into Persian, either manually or via automated methods with post-editing, to ensure linguistic fidelity and idiomatic correctness. This step enabled cross-lingual bias evaluation, sociolinguistic

alignment, and fairness diagnostics in Persian-specific and multilingual LLMs, preserving the social semantics of the original English dataset.

This refined pipeline and dataset provide a robust foundation for quantitative bias assessment, supporting subsequent experiments to evaluate and compare stereotypical tendencies in LLMs across English and Persian contexts.

3.5 Evaluation Metrics

The bias evaluation is conducted using three key metrics:

3.5.1 Sentence Pseudo-Likelihood (PL) Score

Large Language Models (LLMs) like PersianLLaMA and LLaMA generate text based on probability distributions over token sequences, making pseudo-likelihood (PL) scores an effective metric for quantifying bias in sentence completions. Unlike masked language models (MLMs) such as BERT, which predict tokens bidirectionally using masked contexts, PersianLLaMA (13B parameters) and LLaMA (3.1-8B) are autoregressive models, trained to predict the next token given prior tokens in a left-to-right sequence (Touvron et al., 2023). For Persian, a right-to-left (RTL) script language, this left-to-right refers to the logical token order processed by the model after tokenization, not the visual RTL rendering seen by users. Adapted from the StereoSet framework (Nadeem et al., 2021), PL scores assess bias by measuring the likelihood of stereotypical, anti-stereotypical, or neutral completions in both models.

For autoregressive LLMs, the PL score of a sentence $W = w_1, w_2, \dots, w_{|W|}$ is computed as:

$$\text{PL}(W) = \prod_{t=1}^{|W|} P(w_t \mid w_1, w_2, \dots, w_{t-1}) \quad (3.)$$

where $P(w_t \mid w_{1:t-1})$ is the model’s probability of predicting token w_t given the prior context $w_{1:t-1}$, reflecting the autoregressive generation process. PersianLLaMA’s custom tokenizer preprocesses RTL Persian text into a sequential token order that both models handle left-to-right internally, ensuring consistency despite the script’s visual direction (Saffari et al., 2024). For example, a prompt like 'مهندس است' ('Is engineer...') is tokenized and

scored for completions such as 'او کار می کند' ('he/she works') vs. 'او خانه دار است' ('he/she is a homemaker'). Higher PL scores for stereotypical completions (e.g., 'کار می کند' implying a male engineer due to training data bias) over neutral or anti-stereotypical ones (e.g., 'خانه دار است') indicate stronger bias encoding. These scores, applied to StereoSet and the custom template dataset, enable statistical comparisons of bias across PersianLLaMA and LLaMA, supporting the study’s evaluation and mitigation goals.“

3.5.2 Categorical Bias (CB) Score

To aggregate bias across multiple prompts, this study introduces a novel Categorical Bias (CB) score. Building on pseudo-likelihood (PL) scores from prior work (Nadeem et al., 2021; Sap et al., 2020), CB uses variance to capture bias intensity beyond simple preference counts, enabling a comprehensive comparison of PersianLLaMA and LLaMA. It is computed as:

$$CB = \frac{1}{|T|} \sum_{t \in T} \text{Var}_{n \in N}(\log PL_n(t)) \quad (3.)$$

Here, T is the set of prompts from our dataset, with $|T|$ as the total number of prompts. N represents completion types (stereotypical, neutral, anti-stereotypical). For a prompt t , $PL_n(t) = PL(W_n)$ is the PL score of completion W_n of type n . The variance $\text{Var}_{n \in N}(\log PL_n(t))$ measures bias dispersion per prompt; higher variance indicates stronger bias. CB averages these over all prompts, where higher scores reflect greater bias, supporting bias comparison and mitigation testing.

3.5.3 Perplexity-Based Bias Evaluation

In addition to PL Score and categorical bias (CB) scores, perplexity (PPX) was employed as a supplementary metric to assess the implicit preference of language models when presented with stereotype-laden prompts. Perplexity, widely used in language modeling, quantifies how confidently a model assigns probabilities to a sequence of tokens. Lower perplexity indicates greater model confidence, whereas higher perplexity signals uncertainty or implausibility according to the model’s learned distribution (Kurita et al., 2019; Nadeem et al., 2021).

For each StereoSet-style prompt (consisting of stereotype, anti-stereotype, and unrelated completions), we compute the normalized perplexity values

over all three completions using the following formulation:

$$PPX(x) = \exp \left(-\frac{1}{|x|} \sum_{i=1}^{|x|} \log P(x_i \mid x_{<i}) \right) \quad (3.)$$

Where:

- $\mathbf{x} = (x_1, \dots, x_n)$ denotes the tokenized sentence,
- $P(x_i \mid x_{<i})$ is the conditional probability of the i -th token given the preceding tokens $x_{<i} \equiv (x_1, \dots, x_{i-1})$ in the left-to-right generation process.

Perplexity is calculated independently for each sentence and normalized by token length to mitigate sentence-size bias. A model is considered biased if it consistently assigns lower PPX to stereotypical completions, indicating a learned preference for stereotype-conforming text (Sheng et al., 2019).

To ensure fair evaluation across models, we applied architecture-aligned tokenization schemes:

- LLaMA: Byte-Pair Encoding (BPE),
- PersianLLaMA: SentencePiece, optimized for Persian morphology and script.

To avoid tokenizer-induced bias distortions (especially in morphologically rich Persian prompts) we preprocessed all inputs using Unicode normalization, filtered malformed completions, and standardized sentence structures.

Raw perplexity scores can be sensitive to token frequency and language-specific training artifacts, especially in morphologically rich or low-resource languages like Persian. For example, common gendered names or culturally frequent phrases may consistently receive lower perplexity scores regardless of underlying bias (Kaneko et al., 2023). To mitigate this, we compute Mean Group Perplexity and Mean Total Perplexity, which allow us to normalize individual PPX values. Following Siddique et al. (2024), we adopt the Adjusted Perplexity Score (APX), which adjusts for imbalances in group-level distribution by rescaling PPX relative to model and group-specific norms. This yields a fairer and more interpretable cross-model comparison of implicit bias tendencies.

Group-Wise and Overall Perplexity Analysis

To better understand how perplexity patterns differ across social domains, we calculated Mean Group Perplexity, which aggregates PPX over all prompts within a specific bias category (e.g., gender, profession, religion). This revealed which categories elicited stronger model preferences toward stereotypes.

We also computed Mean Total Perplexity, an average across the entire dataset, to capture each model’s overall language confidence and its treatment of biased completions in aggregate.

This two-level perspective (both domain-specific and global) enables comparative profiling of bias trends between LLaMA and PersianLLaMA across English and Persian contexts.

Adjusted Perplexity Score (APX)

To summarize the model’s comparative preference between stereotype and anti-stereotype completions, we adapted the Adjusted Perplexity Score (APX), which evaluates the directional gap between paired completions. APX for a prompt triplet is defined as:

$$APX = \frac{PPX_{\text{anti}} - PPX_{\text{stereo}}}{PPX_{\text{stereo}}} \quad (3.)$$

Where PPX_{stereo} and PPX_{anti} are the normalized perplexities of the stereotype and anti-stereotype completions, respectively.

- A positive APX implies that the model assigns lower perplexity (i.e., greater confidence) to the stereotype, indicating bias.
- A negative APX suggests the model favors the anti-stereotype.
- APX close to zero reflects balanced treatment of both completions.

If a model consistently assigns lower perplexity to stereotypical completions, it indicates that these completions align more closely with the model’s internal language distribution. This implies a learned preference for stereotype-conforming content. Conversely, higher perplexity on anti-stereotypical completions suggests they are treated as less plausible or atypical. Therefore, persistent PPX disparities across domains reveal an implicit bias encoded in the model’s generative behavior.

We compute APX per prompt and then aggregate by domain to identify systemic patterns of stereotype preference. This metric provides an interpretable and quantitative summary of implicit bias across languages and social domains.

3.6 Applications and Model Contextualization

3.6.1 Model Training and Linguistic Scope

PersianLLaMA, a family of language models specifically designed for Persian language processing, was trained on a large-scale, curated corpus of Persian text, encompassing diverse sources such as Persian literature, news articles, social media, and web-based content. This dataset, comprising billions of tokens, was preprocessed to account for Persian’s unique morphosyntactic features, including its right-to-left script and complex verbal morphology. The training objective prioritized optimizing for tasks such as text generation, sentiment analysis, and named entity recognition in Persian, leveraging a SentencePiece tokenizer with Byte Pair Encoding (BPE) to effectively handle Persian’s orthographic and morphological complexities. This tailored approach ensures robust performance across Persian-specific natural language processing (NLP) tasks, making PersianLLaMA particularly suited for applications requiring high linguistic fidelity in Persian contexts.

In contrast, Meta’s LLaMA 3.1, a multilingual model, was trained on a predominantly English-centric corpus, augmented with datasets from multiple languages to support broader multilingual capabilities. While the exact composition of its training data is proprietary, it includes web-scale text, books, and other publicly available sources, with a focus on optimizing for English and other high-resource languages. The model employs a tokenization strategy optimized for Latin-script languages, using a variant of BPE that prioritizes efficiency in English text processing. However, this approach is less effective for languages with non-Latin scripts, such as Persian, where tokenization may fragment words into suboptimal subword units, leading to reduced performance in Persian-specific tasks. Consequently, LLaMA 3.1 serves as a strong baseline for multilingual applications but is less specialized for Persian compared to PersianLLaMA.

3.6.2 Applications of the Custom-Dataset

The Custom-Dataset (inherited from the Steroset dataset), with its bilingual structure and bias-focused design, enables a range of applications in NLP

research and development: Fine-tuning and Evaluating Persian Language Models where the dataset provides a standardized framework for assessing and mitigating representational biases in Persian-specialized models like PersianLLaMA, ensuring fairness in model outputs as well as Designing Fairness-Aware NLP Systems by offering bilingual prompts and completions. The Custom-Dataset supports the development of NLP systems that prioritize equitable representation across cultural and linguistic contexts. Another application of the employed dataset is conducting Cross-Cultural Stereotype Studies. The dataset’s cross-lingual design facilitates comparative analyses of bias manifestation in Persian and English, enabling insights into sociolinguistic variations in stereotype expression.

3.6.3 Tokenization Insights

To illustrate tokenization differences, consider the following examples:

English: “This is the test sentence”

LLaMA 3.1 Tokens: ['This', 'Ġis', 'Ġthe', 'Ġtest', 'Ġsentence']

PersianLLaMA Tokens: ['This', 'Ġ_is', 'Ġ_the', 'Ġ_test', 'Ġ_sentence']

→ Both models tokenize English with clean boundaries, using space-aware prefixes (Ġ for LLaMA, and Ġ for PersianLLaMA via SentencePiece).

Persian: این یک جمله آزمایشی است

LLaMA 3.1 Tokens: *uninterpretable Unicode fragments due to encoding mismatch*

PersianLLaMA Tokens: ['این', 'یک', 'جمله', 'آزمایشی', 'است']

→ PersianLLaMA accurately segments Persian into meaningful subwords aligned with its morphology. LLaMA 3.1, however, produces corrupted byte-level outputs due to tokenizer incompatibility with Persian script and lack of training on Persian Unicode patterns.

These differences in tokenization are not merely cosmetic; they directly influence how effectively each model encodes and generates Persian text. PersianLLaMA’s SentencePiece-based BPE tokenizer captures subword units that reflect Persian’s morphological and syntactic structure, resulting in coherent linguistic representations (Rust et al., 2021). In contrast, LLaMA 3.1’s English-optimized tokenizer often fragments Persian words into inconsistent or semantically meaningless units, degrading contextual understanding (Rust et al., 2021). Consequently, PersianLLaMA demonstrates

superior alignment with Persian-language inputs, while LLaMA 3.1 (though viable for cross-lingual baselining) is not well suited for Persian-specific NLP tasks.

3.7 Analysis & Expected Outcomes

The analysis leverages multiple complementary metrics to characterize how LLaMA and PersianLLaMA internalize and express bias when exposed to stereotype-sensitive inputs in Persian. These metrics include categorical bias (CB), pseudo-likelihood scores (PL), normalized perplexity (PPX), group-level mean perplexity, and the Adjusted Perplexity Score (APX). Each is used to evaluate different dimensions of generative preference and model confidence under bias conditions.

CB scores capture the model’s categorical selection tendencies across stereotype, anti-stereotype, and unrelated completions, offering a discrete view of stereotypical alignment. However, this metric alone does not account for distributional confidence. PL scores, derived from token-level conditional probabilities under masked or autoregressive decoding, provide a more granular measure of how likely a model is to generate each completion under realistic generation dynamics. In our experiments, we use PL to trace preference directionality in contexts where subtle token-level priors can shift the generation away from or toward bias.

To quantify these preferences across domains and normalize for sentence length and structural variation, we compute PPX at the sentence level. This is followed by group-wise and total mean perplexity calculations, which allow us to assess the consistency and variability of model confidence across stereotype categories. APX further refines this by controlling for group-level perplexity variance, producing a normalized score that isolates bias magnitude while accounting for overall model uncertainty.

By analyzing the interaction between these metrics, particularly how PL and PPX align or diverge from CB-based judgments, we aim to determine the extent to which PersianLLaMA’s training regime suppresses or amplifies specific bias patterns. Divergences between PL and CB scores may signal implicit bias encoded at the likelihood level even when categorical outcomes appear neutral. Similarly, shifts in APX distributions across domains provide insight into whether observed biases are domain-specific artifacts or systemic across the model’s linguistic space.

The objective is not only to establish whether PersianLLaMA outperforms LLaMA in mitigating bias, but also to expose how architectural choices, tokenizer design, and corpus-specific pretraining contribute to model behavior. The outcome will inform evaluation design for future fairness-centered LLMs, particularly those targeting morphologically rich, under-resourced languages.

4

Results

Figure 4.1 presents a comparative analysis of sentence length distributions in the custom dataset, contrasting English (blue) and Persian (yellow) across two metrics: character count (left) and word count (right). The left histogram reveals that Persian sentences typically exhibit a broader range of character lengths (peaking around 40–60 characters) compared to English (peaking around 60–80 characters), reflecting Persian's more compact orthography due to its script and lack of vowel diacritics in standard writing.

Conversely, the right histogram shows that English sentences have a higher word count (peaking at 10–12 words) than Persian (peaking at 7–9 words),

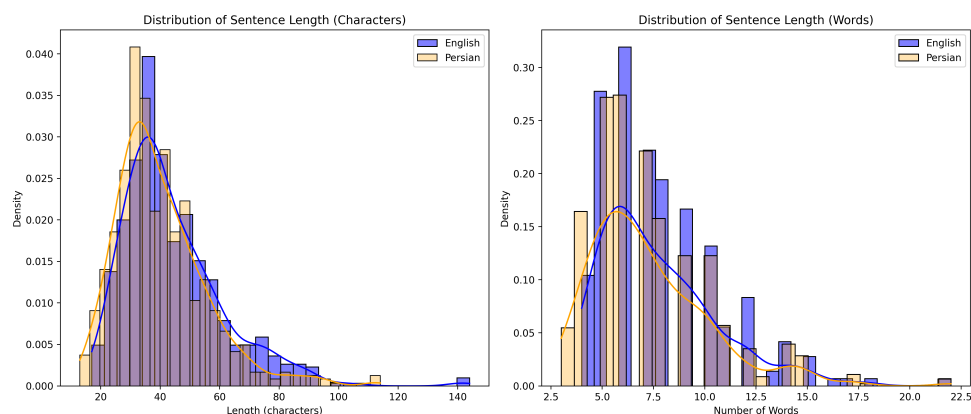


Figure 4.1: Comparative Distribution of Sentence Lengths in the Dataset

likely due to Persian's agglutinative morphology, which often combines multiple semantic units into single words, reducing overall word count. These distributions underscore the linguistic disparities between English and Persian, highlighting the necessity of tailored preprocessing and tokenization strategies, such as SentencePiece with BPE for Persian, to ensure accurate bias evaluation in cross-lingual settings, thereby supporting the development of culturally aware AI systems.

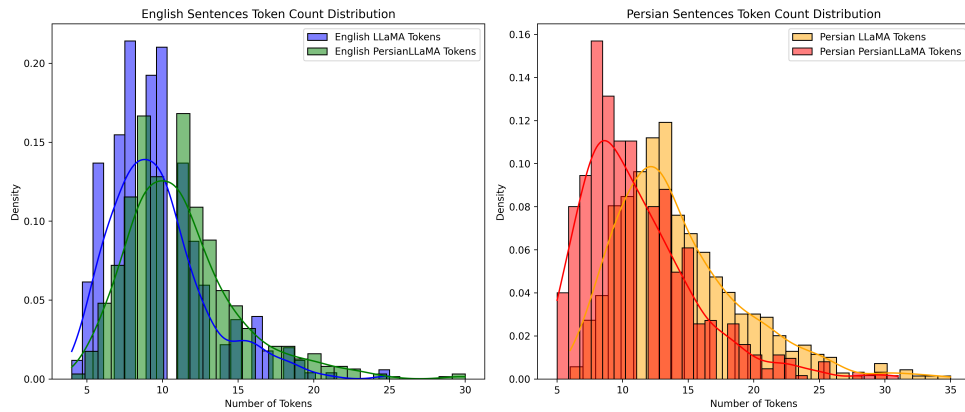


Figure 4.2: Comparative Tokenization Analysis of LLaMA and PersianLLaMA Across English and Persian Sentences

Figure 4.2 presents a comparative analysis of token count distributions for English (left) and Persian (right) sentences from the custom dataset, contrasting the tokenization behavior of LLaMA and PersianLLaMA. In English, both models tokenize with similar distributions, though PersianLLaMA (green) shows a slightly wider spread, peaking around 10–15 tokens compared to LLaMA's peak at 8–12 tokens. In Persian, however, the differences are more pronounced: the orange-yellow bars and red curve (LLaMA) begin later—around 10 tokens—and peak closer to 15–18 tokens, producing longer token sequences for the same sentences. This rightward shift indicates that LLaMA over-segments Persian text due to suboptimal handling of Persian script. By contrast, PersianLLaMA (yellow) yields a concentrated distribution around 10–15 tokens, reflecting more natural segmentation aligned with Persian morphology. These results highlight PersianLLaMA's superior alignment with Persian writing systems and LLaMA's limitations when applied to non-English, morphologically rich languages.

Token length serves as an indicator of how effectively a tokenizer respects linguistic units, particularly in morphologically rich languages like Persian. Well-aligned tokenization produces coherent subword units that correspond

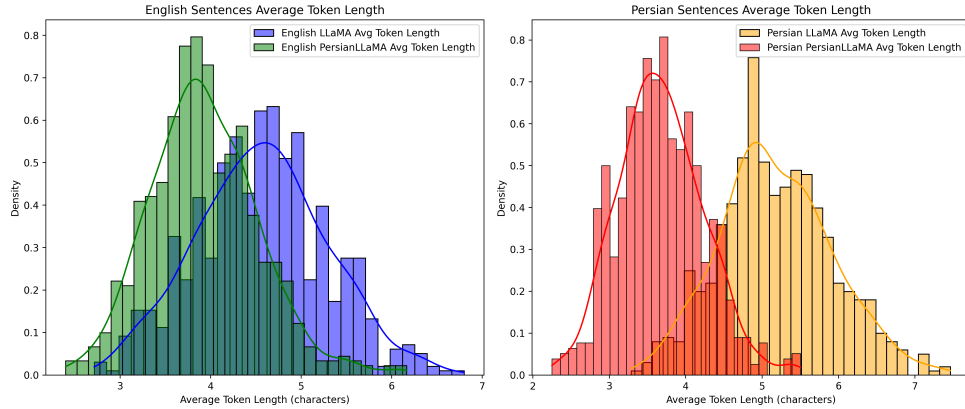


Figure 4.3: Comparative Analysis of Average Token Length Across LLaMA and PersianLLaMA for English and Persian Sentences

to morphemes or whole words, reducing unnecessary fragmentation. Poor tokenization, by contrast, generates overly short and disjointed tokens, inflating sequence length and diluting semantic fidelity. This distinction is evident in the comparative results: PersianLLaMA consistently segments Persian sentences into fewer, more meaningful units, averaging 5–6 characters per token, whereas LLaMA peaks around 4–5 characters, reflecting over-segmentation. Over-segmentation occurs when a tokenizer splits text into too many small pieces because it fails to capture natural Persian word boundaries, introducing noise and undermining downstream interpretability. A similar, though less pronounced, trend is observed in English, where PersianLLaMA again produces slightly longer and more coherent tokens.

Taken together, these findings demonstrate that PersianLLaMA respects sentence structure in Persian and preserves linguistic integrity, whereas LLaMA’s English-optimized BPE vocabulary struggles with Persian, leading to fragmented representations.

These trends are visualized in Figure 4.3, which compares average token length distributions across both languages. When interpreted alongside Figure 4.1 (sentence length in characters) and Figure 4.2 (token count per sentence), a coherent pattern emerges. In Figure 4.1, Persian sentences naturally contain more characters per sentence than English, yet PersianLLaMA maintains shorter overall token counts (Figure 4.2) by grouping characters into longer subword units (Figure 4.3). This triadic relationship highlights

PersianLLaMA’s tokenizer as both morphologically aware and sequence-efficient. In contrast, LLaMA fragments the same high-character Persian inputs into disproportionately more tokens, inflating sequence length (Figure 4.2) while yielding shorter, less meaningful tokens (Figure 4.3).

In the context of bias evaluation, this efficiency is nontrivial: longer, semantically grounded tokens stabilize log-probability estimates and improve the interpretability of PLL and CB scores. Fragmentation in LLaMA introduces spurious variability into log-probability calculations by breaking semantic continuity at the subword level. As such, PersianLLaMA’s tokenization not only enhances morphological alignment but also supports more reliable and linguistically faithful bias quantification in cross-lingual settings.

4.1 Sentence-Level Confidence and Bias Profiling via PLL and CB Metrics

4.1.1 Distribution of Pseudo-Log-Likelihood Scores

Figure 4.4 visualizes the distribution of Pseudo-Log-Likelihood (PLL) scores across English and Persian completions generated by LLaMA and PersianLLaMA. PLL values serve as a normalized measure of how confidently a model assigns probabilities to full sentence completions.

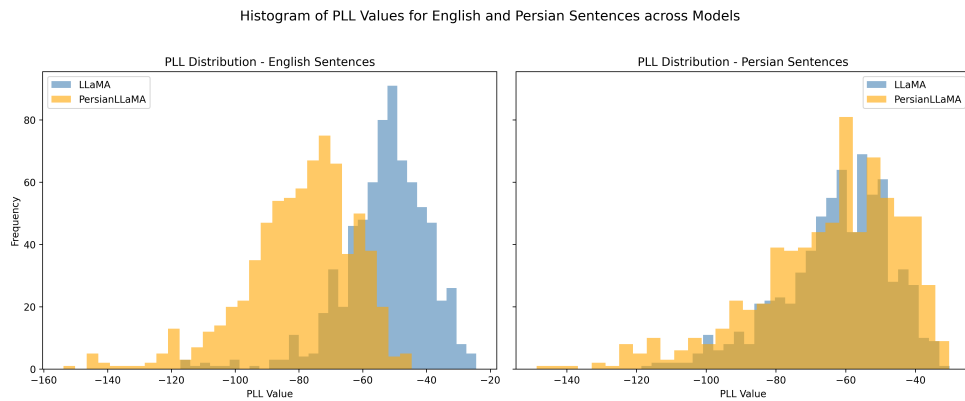


Figure 4.4: Distribution of Pseudo-Log-Likelihood (PLL) Values for English and Persian Sentences in LLaMA and PersianLLaMA Models

For English completions (Figure 4.4, left), LLaMA shows a tighter and higher-centered distribution of PLL values—predominantly between -60 and -40; consistent with its English-centric training corpus. PersianLLaMA, while still functional in English, produces a broader PLL distribution that skews

slightly lower (centered near -70), indicating lower fluency and model uncertainty in non-target language contexts.

In contrast, Persian sentences (Figure 4.4, right) yield an inverse pattern: PersianLLaMA displays a more compact distribution with peak PLL values between -60 and -45, while LLaMA demonstrates a flatter spread extending toward -100. This contrast reflects PersianLLaMA’s tokenizer and pretraining corpus alignment with Persian morphology, enabling more confident generation in that language. The asymmetry underscores a language specialization trade-off: LLaMA generalizes well in English but underperforms on morphologically rich scripts like Persian, whereas PersianLLaMA achieves native fluency at the cost of degraded cross-lingual generalization.

4.1.2 Pairwise PLL Correlation Between Models

The sentence-level scatter plots in Figure 4.5 compare PLL scores from both models for each sentence in the dataset. The English comparison (left) exhibits a strong positive correlation, with most points aligned along the diagonal. This indicates consistent sentence-level scoring behavior across models, though LLaMA tends to assign higher PLL values overall.

The Persian comparison (right) shows a greater dispersion. PersianLLaMA systematically assigns higher PLL values to Persian completions, clustering closer to -50, while LLaMA’s scores are frequently lower, sometimes approaching -100. The gap between model confidence levels in Persian is both statistically and practically significant, demonstrating PersianLLaMA’s better internal calibration in its training language.

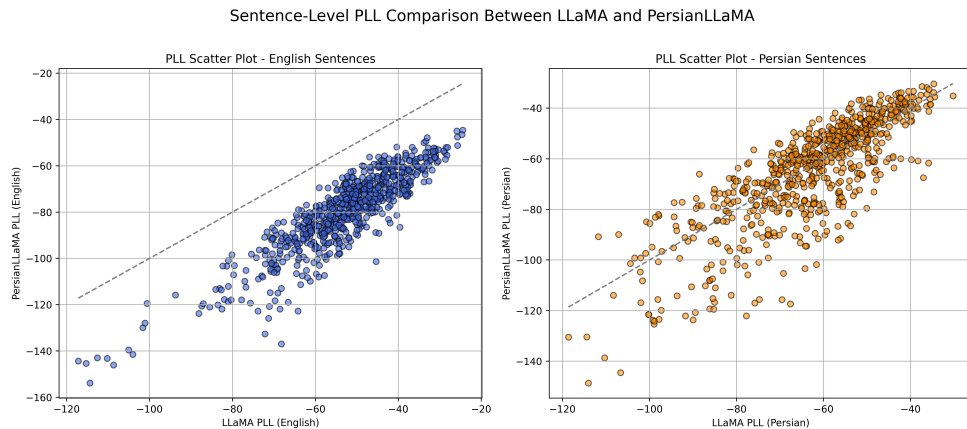


Figure 4.5: Sentence-Level Pseudo-Log-Likelihood (PLL) Correlation Between LLaMA and PersianLLaMA for English and Persian Sentences

This divergence is particularly relevant to fairness evaluation: under-confident or inconsistent scoring (as seen in LLaMA's Persian completions) increases the risk of misinterpreting stereotype–anti-stereotype differentials. Thus, PersianLLaMA provides a more stable substrate for bias diagnostics in Persian contexts.

4.1.3 Bias Score Distribution Across Languages and Models

Figure 4.6 depicts the distribution of PLL-derived bias scores (computed as the difference in PLL between stereotype and anti-stereotype completions) for both models across English and Persian inputs. In English, LLaMA's scores center near zero but are tightly clustered between -20 and 0, indicating a moderate but consistent tendency to prefer stereotype completions. PersianLLaMA's English completions are more dispersed, suggesting higher variance and occasional reversal of preference.

In Persian, PersianLLaMA demonstrates a slightly tighter and more neutral distribution (between -20 and 0) compared to LLaMA, whose outputs span a wider range (from -60 to +20). This reflects both a more calibrated handling of stereotypical phrasing and a smoother bias profile in PersianLLaMA.

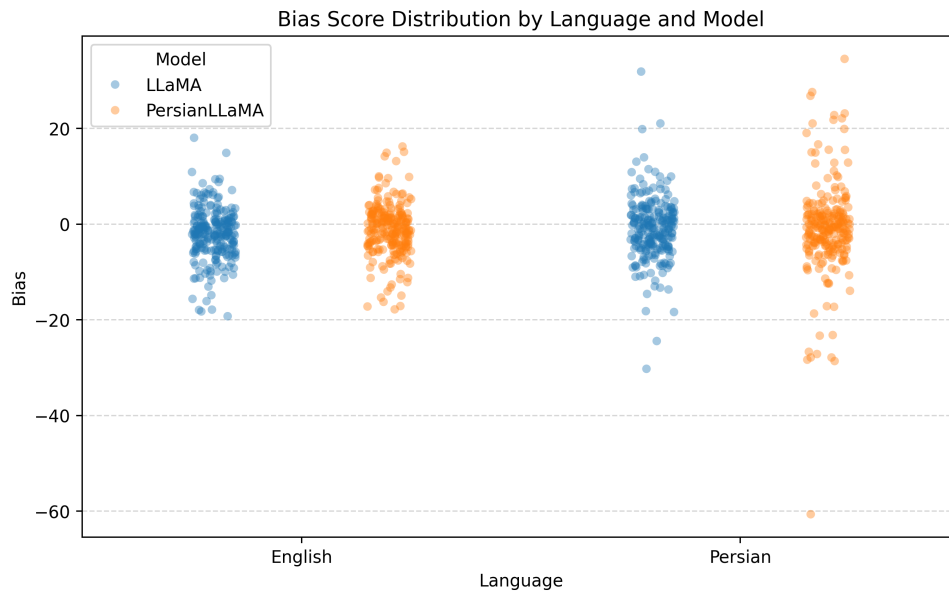


Figure 4.6: Bias Score Distribution of LLaMA and PersianLLaMA Across English and Persian Languages in the dataset

Taken together, this comparison validates the methodological assumption that PLL gaps serve as directional bias indicators. PersianLLaMA’s compact Persian-specific distribution suggests it maintains representational fairness more effectively in its primary language, while LLaMA’s noisy output in Persian highlights the dangers of evaluating bias using multilingual models in unsupported scripts.

4.1.4 Contextual Bias Scores by Demographic Category

Figure 4.7 presents Contextual Bias (CB) scores, derived from the variance in PLL among the three completions, across four bias categories: race, religion, profession, and gender. This metric captures the degree to which a model differentiates among completions for a given prompt.

In English, LLaMA produces moderate CB scores across all categories, ranging from 20–30. Its strongest differentiation appears in religion and gender prompts. PersianLLaMA’s English CB scores remain comparable but occasionally drop below 20, indicating reduced distinction in its non-primary language.

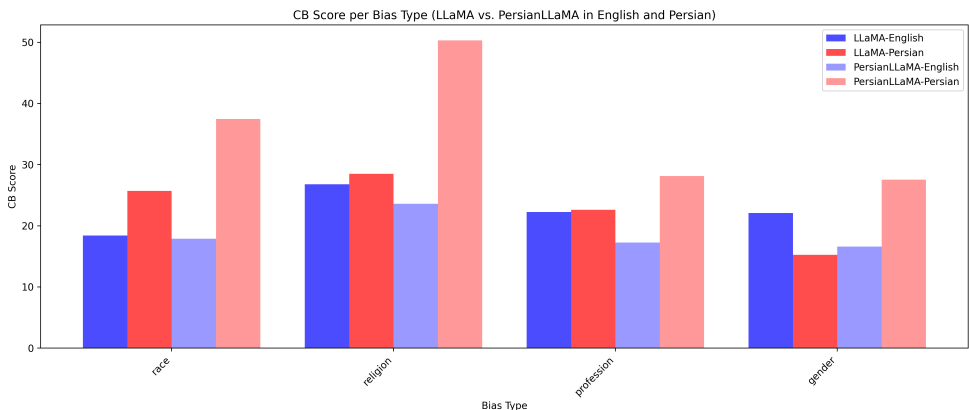


Figure 4.7: Contextual Bias (CB) Scores Across Bias Categories for LLaMA and PersianLLaMA in English and Persian Contexts

For Persian completions, PersianLLaMA’s Contextual Bias (CB) scores vary more distinctly across categories. It registers its highest CB in religion-related prompts (approximately 50), followed by gender (approximately 30), and profession (approximately 28). This distribution may reflect the model’s exposure to culturally encoded distinctions within its Persian training corpus, which enhances sensitivity in certain domains but may also amplify variance specific to those contexts.

Interestingly, LLaMA's Persian CB scores are more uniform but lower overall. While this may suggest less bias, it more likely reflects LLaMA's reduced confidence and linguistic incoherence in Persian, which flattens scoring differentiation due to under-modeling.

4.1.5 Bias Intensity by Label Type and Category

To explore how model confidence varies with respect to stereotype polarity within specific demographic domains, we analyze the mean Pseudo-Log-Likelihood (PLL) scores for each completion label (stereotype, anti-stereotype, unrelated), grouped by bias type (race, gender, religion, profession). Figures 4.8 and 4.9 present these results for English and Persian inputs, respectively.

In Figure 4.8, which focuses on English sentences, LLaMA consistently achieves higher PLL scores (i.e., lower perplexity and greater fluency) than PersianLLaMA across all bias types and label categories. For example, in the religion–stereotype group, LLaMA's mean PLL approaches -50, while PersianLLaMA's mean hovers around -70. This pattern reflects LLaMA's training on English-rich corpora and confirms its strong syntactic alignment and representational confidence in English. Across all four bias types, the PLL gap between stereotype and anti-stereotype completions is consistently larger in LLaMA, with stereotype completions rated more confidently. PersianLLaMA, by contrast, shows weaker overall fluency in English and a reduced ability to differentiate between label types.

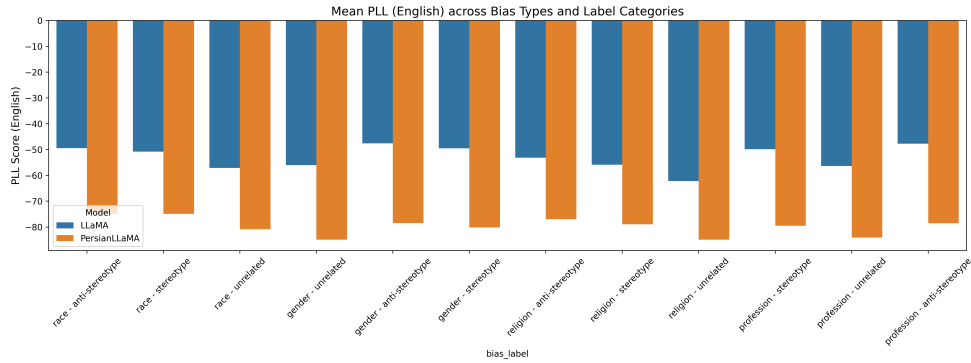


Figure 4.8: Mean Pseudo-Log-Likelihood (PLL) Scores for English Sentences Across Bias Types and Label Categories Using LLaMA and PersianLLaMA

Figure 4.9 shows the same analysis applied to Persian completions, offering a contrasting view of each model's behavior in a morphologically complex, low-resource language. Here, PersianLLaMA exhibits higher PLL scores than

LLaMA across all categories, particularly for stereotype and unrelated completions, confirming its superior alignment with Persian linguistic structure. Notably, the PLL gaps between stereotype and anti-stereotype are smaller in PersianLLaMA, indicating a more balanced treatment of biased and neutral continuations. For instance, in the gender and profession categories, both stereotype and anti-stereotype completions are scored similarly by PersianLLaMA, while LLaMA continues to favor stereotypes with higher PLLs despite its lower fluency overall.

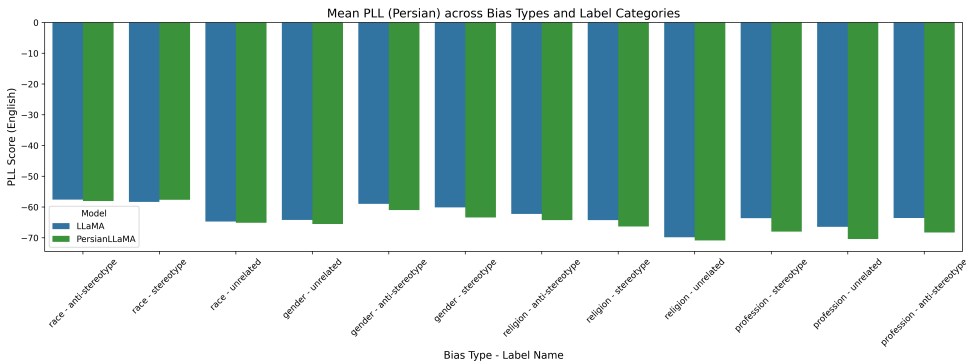


Figure 4.9: Evaluating Language Alignment: Mean PLL Comparison of LLaMA and PersianLLaMA across Bias Categories in Persian

This cross-lingual comparison reveals three key insights. First, LLaMA exhibits stronger directional bias in English, consistently scoring stereotype completions more confidently. Second, PersianLLaMA demonstrates reduced bias polarization in Persian, suggesting that culturally aligned pre-training mitigates bias reinforcement in generative preferences. Third, the PLL gaps in Persian are narrower overall, indicating that PersianLLaMA’s language-specific optimization not only improves fluency but also supports fairer scoring dynamics across demographic dimensions.

Taken together, these findings underscore the importance of linguistic alignment in fairness evaluation. While LLaMA performs strongly in its primary language, it exhibits increased directional bias. PersianLLaMA’s more equitable scoring in Persian illustrates the potential of targeted pretraining for bias-sensitive NLP in underrepresented languages.

4.2 Statistical Analysis of Model Confidence Across Languages

To quantitatively support the observed differences in model behavior, we computed descriptive statistics and conducted paired inferential tests on sentence-level Pseudo-Log-Likelihood (PLL) values across both English and Persian prompts in the dataset. These results complement the distributional insights shown in Figures 4.4 and 4.5 and provide statistical validation for the language-specific strengths observed in each model.

Table 4.1 presents the mean and variance of PLL values for LLaMA and PersianLLaMA across both language contexts. As expected, LLaMA demonstrates higher mean PLL scores (i.e., lower perplexity) in English, consistent with its English-centric training. Conversely, PersianLLaMA achieves marginally better mean scores in Persian, alongside a higher variance; indicating increased expressivity and sensitivity to structural variations in the Persian language.

Table 4.1: Descriptive Statistics for PLL Scores (Sentence-Level)

Metric	LLaMA	PersianLLaMA
Mean PLL (English)	-52.95	-79.76
Variance of PLL (English)	185.86	303.89
Mean PLL (Persian)	-62.81	-64.89
Variance of PLL (Persian)	244.01	436.05

To assess the statistical significance of these differences, we applied paired t-tests on sentence-aligned PLL values for both models. The results are reported in Table 4.2. In English, the large t-statistic (89.65) and near-zero p-value (< 0.0001) confirm that LLaMA significantly outperforms PersianLLaMA in terms of fluency and confidence. In Persian, the t-statistic (4.38) is smaller but still statistically significant ($p < 0.0001$), suggesting that PersianLLaMA holds a measurable advantage when evaluated in its target language.

Table 4.2: Paired t-Test Results for PLL Scores (LLaMA vs. PersianLLaMA)

Language	t-statistic	p-value
English	89.65	< 0.0001
Persian	4.38	< 0.0001

These results confirm that the observed performance trends are not due to random variance but represent robust model-level differences. The significantly higher variance in PersianLLaMA’s Persian completions suggests increased sensitivity to nuanced morphological or cultural variation, reinforcing its appropriateness for bias-sensitive evaluations in underrepresented language settings. Furthermore, the strong statistical significance in both languages strengthens the argument that linguistic alignment is a critical factor in achieving interpretable and fair probabilistic behavior in generative models.

4.3 Perplexity and Adjusted Perplexity Index (APX): Diagnostic Insights into Cross-Lingual Bias Dynamics

To complement PLL-based findings, we incorporate sentence-level perplexity and the derived Adjusted Perplexity Index (APX) as diagnostic tools to assess model fluency and representational bias across identity-sensitive prompts. These metrics provide deeper insights into how LLaMA and PersianLLaMA diverge in handling semantically loaded completions across languages.

4.3.1 Cross-Linguistic Perplexity Trends by Bias Type

Figure 4.10 compares raw perplexity (PPL) scores across four bias domains evaluated across both models in English and Persian. LLaMA maintains consistently low perplexity across categories in English, aligning with its training focus. Its perplexity scores in Persian are also relatively constrained, though slightly elevated, suggesting partial generalization.

In contrast, PersianLLaMA demonstrates highly polarized behavior. In Persian, it achieves relatively low perplexity across all domains, indicating robust modeling in its target language. However, its perplexity on English input (particularly for profession-related prompts) exceeds 28,000, reflecting severe misalignment and decoding instability. This spike underscores a critical limitation: PersianLLaMA lacks the linguistic grounding to fluently process stereotype-laden English content, particularly in domains not reflected in its pretraining corpus.

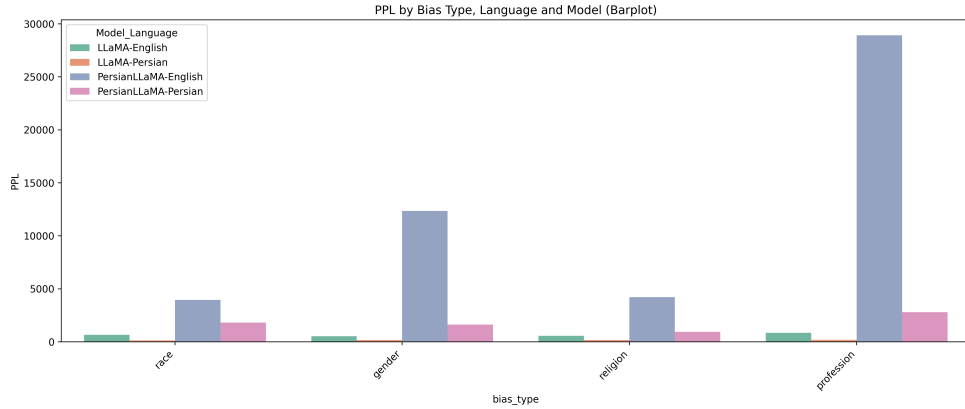


Figure 4.10: Cross-Linguistic Perplexity Analysis by Bias Type: A Comparative Study of LLaMA and PersianLLaMA Performance

4.3.2 Bias Sensitivity via Adjusted Perplexity Index (APX)

Figure 4.11 presents the APX values per bias category. APX captures the relative increase in perplexity for stereotype-sensitive prompts compared to a model’s overall expectations, thereby surfacing identity-specific uncertainty.

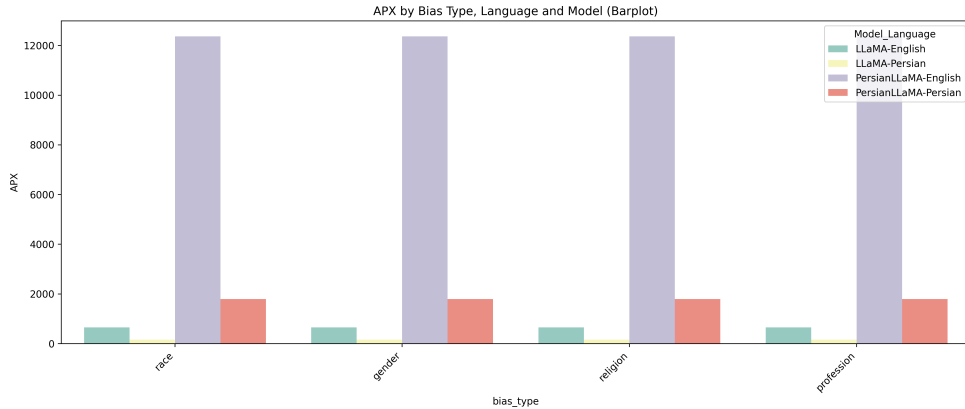


Figure 4.11: Adjusted Perplexity Index (APX) across Bias Domains: Evaluating Cross-Lingual Fairness

PersianLLaMA in English yields extremely high APX values (approximately 12,000), confirming both its domain misalignment and hypersensitivity to unfamiliar social semantics. In its native Persian setting, PersianLLaMA maintains much lower APX values (approximately 1,800), reflecting better calibration to stereotype constructs. LLaMA, meanwhile, displays minimal

APX variation in both English ($\sim 700\text{--}800$) and Persian (~ 100), suggesting stable, albeit linguistically under-expressive, behavior. The near-flat APX profile in Persian suggests that LLaMA under-reacts to demographic distinctions, potentially masking biases rather than confronting them.

4.3.3 Bias Label Divergence in English and Persian

Figures 4.12 and 4.13 break down mean PPL across bias-label combinations (e.g., gender–stereotype, religion–anti-stereotype), providing fine-grained views into how each model navigates specific social representations.

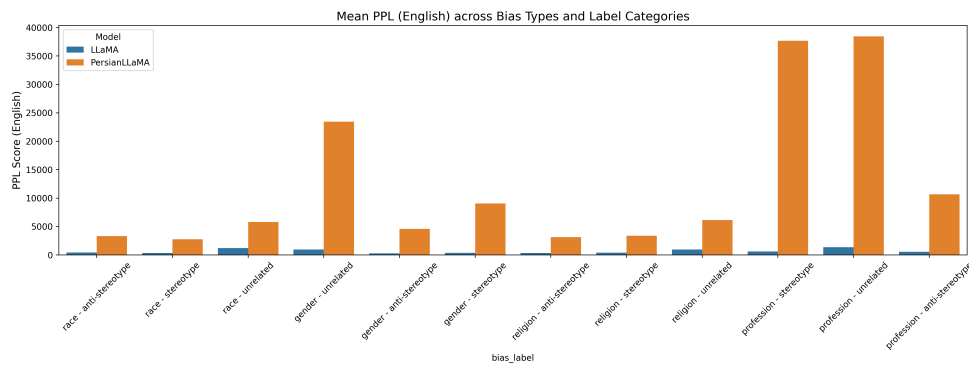


Figure 4.12: Mean PPL in English by Bias Type and Label: Highlighting Model Divergence

In English (Figure 4.12), LLaMA outperforms PersianLLaMA across all combinations, with particularly low perplexity for stereotype completions; indicating its bias-reinforcing tendencies. PersianLLaMA, in contrast, demonstrates significantly elevated perplexity for unrelated and anti-stereotype completions, suggesting not only reduced fluency but also inconsistent treatment of bias polarity in out-of-domain input.

In Persian (Figure 4.13), PersianLLaMA reverses this pattern, outperforming LLaMA across nearly all bias-label groups. It shows variation in perplexity across labels, particularly for *stereotype* and *unrelated* completions, suggesting contextual sensitivity. LLaMA again underperforms in Persian, with nearly flat PPL scores ($\sim 300\text{--}500$) across all categories which is signaling semantic indifference and low discriminatory power.

4.3.4 Cross-Lingual Correlation of Bias Metrics

Figure 4.14 presents the cross-lingual correlation between English and Persian metrics. The APX values exhibit only weak correlations across languages

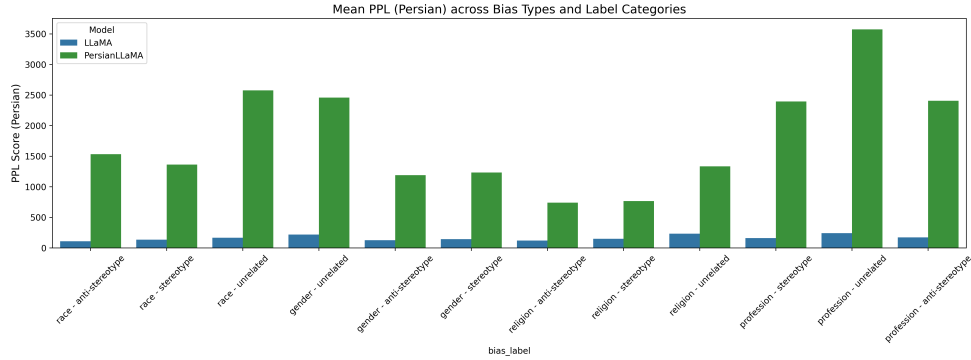


Figure 4.13: Mean PPL in Persian across Bias Types and Label Categories: Evaluating Performance on Native Language Input

for both models, with $r = 0.314$ for PersianLLaMA and $r = 0.258$ for LLaMA. These low correlation coefficients suggest that representational bias does not generalize symmetrically across linguistic boundaries. This finding underscores the importance of language-specific evaluation, as bias observed in one language cannot be assumed to translate reliably into another, particularly when cultural semantics and lexical conventions diverge.

By contrast, sentence-level fluency metrics (PLL) demonstrate stronger cross-lingual alignment. LLaMA exhibits a high correlation between English and Persian PLL scores ($r = 0.748$), with PersianLLaMA following closely ($r = 0.675$). These results indicate that linguistic fluency tends to generalize more reliably across languages than sociolinguistic bias.

English vs Persian: Correlation Analysis by Model and Metric

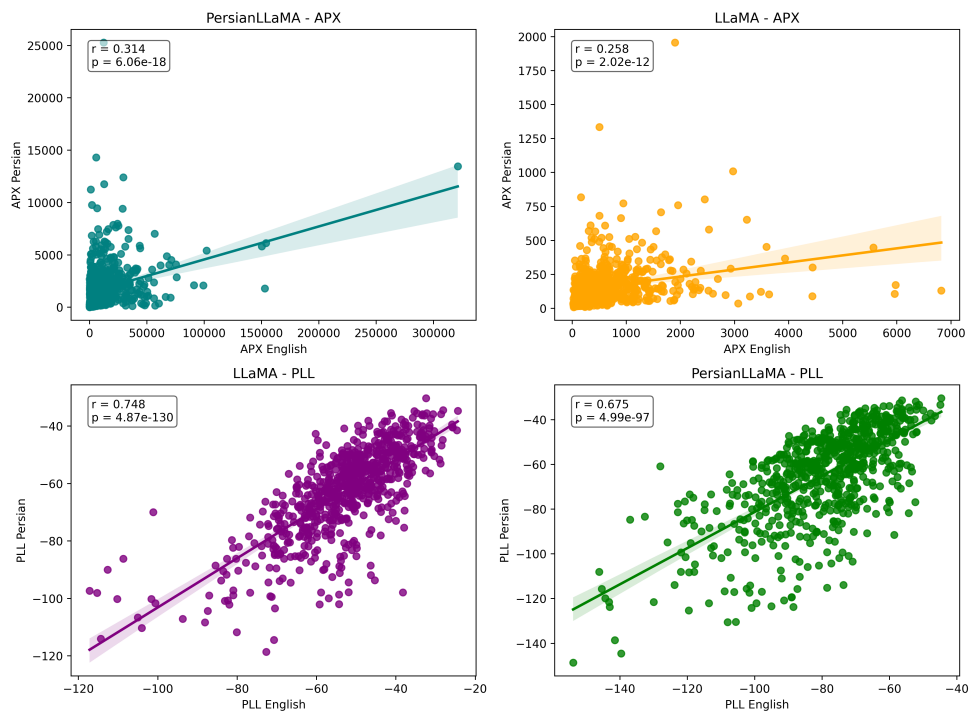


Figure 4.14: Cross-Lingual Consistency in Model Behavior: Correlation of English and Persian Metrics

5

Discussion

5.1 Summary

This study systematically evaluated representational and directional bias in two large-scale language models (LLaMA 3.1–8B-Instruct and PersianLLaMA-13B) across English and Persian, using a controlled cross-lingual variant of the StereoSet benchmark. By combining pseudo-log-likelihood scoring, normalized perplexity, adjusted perplexity index (APX), and contextual bias metrics (CB), the analysis provided a multi-layered quantification of bias across demographic dimensions such as gender, race, religion, and profession.

The findings reveal that LLaMA, pretrained predominantly on English-centric corpora using a BPE tokenizer optimized for Latin scripts, demonstrates high sentence-level fluency and low perplexity in English but struggles with morphological and semantic coherence in Persian. Its PLL values were consistently higher in English, and APX scores remained low, suggesting balanced fluency and bias generalization in high-resource settings. However, in Persian, LLaMA showed flattened PLL and APX distributions, indicating semantic underactivation and a potential insensitivity to socio-cultural variation, a finding consistent with prior evaluations of multilingual models on underrepresented languages (Rust et al., 2021; Wang et al., 2024).

PersianLLaMA, on the other hand, pretrained on Persian data using a SentencePiece tokenizer, exhibited strong modeling capacity in its native language. It generated more morphologically consistent token sequences, demonstrated lower perplexity, and displayed higher variance in bias-sensitive prompts, suggesting a nuanced representation of Persian identity constructs. Its adjusted perplexity values in Persian remained bounded, but when evaluated on English prompts (especially profession and religion) the model exhibited severe degradation, including inflated APX values exceeding 12,000, reflecting poor cross-lingual transferability. Despite this, PersianLLaMA maintained consistent CB and PLL distributions in Persian, demonstrating effective domain specialization for low-resource, morphologically rich languages.

The model-level comparison highlighted three distinct behavioral regimes: LLaMA as a generalist with fluent English performance but limited expressivity in Persian; PersianLLaMA as a specialist with calibrated sensitivity in its native language but high instability when applied to unseen scripts and sociolinguistic contexts; and the inadequacy of existing fluency metrics to fully capture representational bias in cross-lingual settings. Furthermore, correlation analysis between English and Persian APX and PLL scores suggested that sentence-level fluency generalizes more reliably across languages than sociocultural bias patterns, which were shown to be highly language-specific.

This evaluation offers new empirical grounding for claims that language-specific pretraining yields deeper semantic modeling and finer-grained bias differentiation in low-resource contexts, a hypothesis previously underexplored in fairness-focused LLM research. It also reinforces the methodological principle that multilingual bias audits must disaggregate model behavior by language, bias category, and evaluation metric. The hybrid use of PLL, perplexity, and APX in this study enables a granular understanding of how large language models encode, amplify, or suppress stereotypical associations across demographic and linguistic boundaries.

5.2 Analysis and Interpretation

The results presented in this study illuminate the complex interplay between linguistic alignment, model architecture, and bias propagation in large language models. By dissecting the behavior of LLaMA and PersianLLaMA across multiple evaluation metrics and languages, we demonstrate that

representational bias is not a monolithic property but rather a dynamic function of tokenization granularity, pretraining corpus composition, and language-specific morphological structure.

LLaMA’s superior performance in English, evidenced by higher PLL scores, lower perplexity, and narrow adjusted perplexity index (APX) distributions, aligns with its pretraining on predominantly English corpora. The model exhibits a consistent directional preference for stereotype completions in English, particularly within religion and gender domains. This is reflected in higher contextual bias (CB) scores and wider log-probability margins between stereotype and anti-stereotype completions. Importantly, LLaMA’s low APX variance suggests a well-calibrated treatment of demographic groups in fluent English contexts, yet its performance in Persian remains limited. The model’s flattened perplexity and CB profiles in Persian indicate a semantic plateau, a phenomenon frequently associated with under-tokenization or lack of morphological alignment in low-resource scripts (Wang et al., 2024).

PersianLLaMA, by contrast, excels in Persian due to its domain-specific pretraining and the use of a subword vocabulary designed to preserve morphological coherence. Its lower perplexity and higher variance in PLL and CB metrics suggest not only fluency but also semantic differentiation aligned with Persian social structures. Notably, PersianLLaMA maintains narrower PLL gaps between stereotype and anti-stereotype completions, pointing to reduced directional bias. However, this strength does not generalize: when evaluated on English prompts, the model’s performance deteriorates significantly, with inflated perplexity values and extreme APX scores, particularly in profession- and religion-related categories. This instability underlines a critical limitation of language-specialized models, they are highly sensitive to domain drift and unable to generalize semantic bias representations beyond their training language.

The contrast between models is particularly evident in their tokenization behavior. PersianLLaMA’s SentencePiece tokenizer yields longer, semantically grounded tokens in Persian, which improves representational clarity. LLaMA’s BPE tokenizer, optimized for English, performs well in its native language but over-fragments Persian text, degrading contextual fluency and distorting probabilistic scoring. These tokenizer-induced discrepancies propagate downstream, affecting CB and APX measurements. Our findings reinforce that tokenization is not merely a preprocessing step, but a central

Table 5.1: Model Comparison Analysis Table (All numeric values are approximate)

Metric/ Behavior	LLaMA (EN)	LLaMA (FA)	P LLaMA (EN)	P LLaMA (FA)	Interpretation
Mean PLL	High (−52.95)	Moderate (−62.81)	Low (−79.76)	High (−64.89)	Fluency matches training language
CB Score	High (30–35)	Low (20–25)	Low–Moderate (25–30)	Moderate (30–50)	Higher discrimination in native language
APX	Low (700–800)	Very Low (100)	Very High (12,000–13,000)	Moderate (1,800)	Bias less detectable out-of-domain
PLL–APX Corr.	Strong ($r = 0.75$)	Weak ($r = 0.25$)	Moderate ($r = 0.31$)	Moderate ($r = 0.31$)	Fluency generalizes better than bias sensitivity
Tokenization	Good	Poor	Poor	Excellent	Drives fluency and score stability
Bias Sensitivity	Gender, Religion	Flat	Profession, Religion	Gender, Religion	Domain-specific divergence

architectural factor in how models internalize sociolinguistic features, as supported by Rust et al. (2021) and Blevins and Zettlemoyer (2022).

The cross-lingual correlation analyses further illuminate these dynamics. PLL scores, which quantify sentence-level fluency, exhibit strong alignment across languages ($r > 0.67$), indicating that surface-level coherence generalizes relatively well between English and Persian. In contrast, APX, capturing perplexity deviations driven by bias sensitivity, shows weak cross-lingual correlation ($r = 0.25$ – 0.31), suggesting that representational bias does not transfer symmetrically across linguistic contexts. This divergence underscores a critical theoretical insight: while fluency appears to generalize across languages, bias remains highly language-specific, necessitating localized evaluation frameworks.

In sum, the analysis reveals a methodological imperative. Cross-lingual fairness cannot be inferred from monolingual metrics or evaluated solely on high-resource languages. Instead, bias evaluations must incorporate language-specific scoring, culturally relevant prompts, and comparative metrics like APX and CB. The current findings suggest that multilingual

LLMs require both broader data coverage and finer language-aware calibration to achieve equitable performance across diverse linguistic populations.

To complement the preceding analysis, Table 5.1 summarizes the core evaluation metrics. It highlights how each model performs under sociolinguistic stress, confirming that PersianLLaMA offers superior alignment for Persian while LLaMA remains dominant in English. These findings reinforce that language-specific pretraining not only enhances linguistic fidelity but also plays a critical role in accurately modeling sociocultural nuances relevant to bias evaluation.

5.3 Limitations

While this study offers a comprehensive, architecture-aware evaluation of bias in large language models (LLMs) across English and Persian, several methodological and interpretive limitations should be acknowledged.

First, the comparative evaluation hinges on the StereoSet benchmark, which, despite being a widely adopted dataset for stereotype detection, was originally designed for English and adapted here for Persian through translation. Although meticulous care was taken to preserve semantic equivalence and cultural relevance, the translated prompts may still carry linguistic artifacts or introduce subtle meaning shifts that influence model behavior. This poses a challenge in guaranteeing that the stereotype-anti-stereotype distinction retains the same sociolinguistic salience across languages.

Second, the study evaluates only two models; LLaMA-3.1-8B-Instruct and PersianLLaMA-13B. Although these choices represent a balanced pairing between a multilingual baseline and a language-specific model, broader generalization would require inclusion of additional architectures, such as mBERT, XGLM, or DeepSeek-Coder. Furthermore, the models were evaluated in 4-bit quantized format to ensure computational feasibility. While quantization is known to preserve core semantic behaviors, it may introduce subtle distortions in confidence estimates, particularly in the form of log-probability (PLL) scores and derived metrics like APX and CB.

Another constraint lies in the use of perplexity-based metrics as proxies for bias. Although widely adopted in prior work (Nadeem et al., 2021; Siddique et al., 2024), these metrics assume that confidence gaps between stereotype

and anti-stereotype completions reliably encode bias directionality. However, perplexity alone may conflate linguistic fluency with social preference, especially in morphologically complex languages like Persian where sub-word segmentation impacts token frequency and distributional likelihood. While our use of APX aimed to normalize such effects, future work could incorporate more causal or contrastive methods (e.g., counterfactual perturbation or intervention-based diagnostics) to more precisely isolate bias encoding.

The imbalance in training data coverage also presents a systemic limitation. LLaMA’s superior performance in English stems from its pretraining on a large and diverse corpus of English-centric texts, while its lower fidelity in Persian reflects both tokenization mismatches and limited exposure to Persian content. PersianLLaMA, while more effective in native contexts, demonstrates volatility in cross-lingual generalization, particularly in domains like profession and religion. However, the pretraining data for PersianLLaMA is not publicly disclosed in full, which limits interpretability of its learned biases.

Finally, statistical testing in this study relied on sentence-level paired t-tests. While these support the significance of observed differences, they do not fully capture contextual dependencies across tokens or prompt structures. More robust statistical modeling (such as mixed-effects regression or Bayesian calibration) could provide finer-grained inference across categories and better control for prompt-level variability.

Despite these limitations, the evaluation framework remains valid and informative. It highlights not only the architectural and linguistic distinctions between LLMs, but also underscores the need for culturally calibrated evaluation methodologies that go beyond English-dominant paradigms. These limitations offer a concrete basis for refining future multilingual bias assessments in LLMs.

5.4 Ethical and Societal Consequences

The findings of this thesis have direct ethical and societal implications, particularly because biased language models can reinforce or amplify harmful stereotypes related to gender, religion, race, and occupation. As prior work has shown, biased model outputs risk marginalizing already underrepresented groups and may negatively influence downstream applications such

as hiring, content moderation, educational systems, or legal decision support (Nadeem et al., 2021; Parrish et al., 2022; Siddique et al., 2024).

In the context of Persian, a low-resource and culturally rich language, biased model behavior may cause disproportionate harm due to the scarcity of alternative tools and benchmarks (Mohammadi et al., 2024). This makes fairness evaluation particularly important, as emphasized in multilingual bias research (Kaneko et al., 2023; Milios & BehnamGhader, 2022). The results of this study highlight that multilingual LLMs not optimized for Persian may produce culturally insensitive or inaccurate responses, which could mislead users or reinforce inaccurate sociocultural associations.

Furthermore, any deployment of Persian-capable LLMs in public-facing applications should consider the structural biases revealed in this work. Developers and policymakers must ensure that LLMs used in Persian-speaking regions are built with linguistically aligned tokenizers, culturally balanced datasets, and explicit fairness safeguards. This reinforces the growing consensus that fairness, sociolinguistic alignment, and responsible evaluation must be treated as primary design requirements for non-English LLMs rather than downstream adjustments.

5.5 Future Research Directions

Building upon the findings of this study, several technically grounded avenues emerge for future research aimed at advancing bias evaluation and mitigation in multilingual language models, particularly those operating in low-resource or typologically diverse languages such as Persian.

First, our results highlight the need for context-aware calibration methods that extend beyond log-probability-based scoring. While metrics like Pseudo Log-Likelihood (PLL), Contextual Bias (CB), and Adjusted Perplexity Index (APX) offer useful signals, they remain sensitive to lexical frequency, sentence length, and tokenization behavior. Future work should integrate semantic similarity metrics (e.g., sentence embeddings or cosine similarity over contextualized representations) to complement perplexity-based bias detection. This could help mitigate spurious confidence differences introduced by surface-form divergence across translations.

Second, despite using carefully aligned prompts in English and Persian, cultural equivalence remains a challenge in cross-lingual stereotype benchmarking. Future research should incorporate culturally adaptive stereotype

generation frameworks, potentially leveraging multilingual crowd - sourcing or grounded sociolinguistic datasets to better align bias typologies with regional and cultural norms. The creation of large-scale, culture-specific datasets with fine-grained bias annotations (e.g., class, ethnicity, dialect) would strengthen the ecological validity of bias evaluation in underrepresented settings.

Third, while this study focused on the causal language modeling task, future extensions could investigate bias propagation in downstream applications, such as named entity recognition, machine translation, or question answering, where bias can be amplified through structured prediction pipelines. Analyzing how representational bias influences system-level behavior would provide broader insight into real-world fairness implications.

Fourth, current models rely heavily on subword tokenization schemes like BPE or SentencePiece, which while effective for compression, may obfuscate underlying linguistic structure. Future architectures might explore morphology-aware tokenizers (e.g., Morfessor or character-based models) for languages like Persian, where affixation and compounding significantly affect semantic interpretation. This could yield fairer representations by minimizing tokenization-induced bias variance.

Finally, a critical research direction involves algorithmic fairness interventions integrated during pretraining or fine-tuning, such as counterfactual data augmentation, bias-constrained loss functions, or adversarial debiasing (Zhang et al., 2023). These techniques should be evaluated not only in English but also in typologically diverse, morphologically rich languages to ensure generalizability. In particular, contrastive fine-tuning using matched stereotype/anti-stereotype pairs in native language corpora may offer a viable path toward equitable multilingual LLMs.

In summary, addressing representational bias in multilingual NLP systems demands innovation in dataset construction, evaluation methodology, model architecture, and training strategies. Future work must prioritize linguistically and culturally grounded fairness to ensure equitable and accountable AI across global language communities.

6

Conclusion

This thesis evaluated how bias manifests in Large Language Models (LLMs) within non-English linguistic settings, using Persian as a representative case. The study addressed two key research questions:

1. Do PersianLLaMA and LLaMA differ in amplifying or mitigating gender, racial, religious, and occupational biases in Persian text processing?
2. How do tokenization strategies and language alignment influence the measurement and manifestation of bias in Persian LLMs?

Regarding the first question, the analysis showed clear differences in bias behavior between PersianLLaMA and LLaMA. PersianLLaMA, trained on Persian corpora and equipped with a SentencePiece tokenizer, exhibited lower contextual and representational bias across all domains. Its narrower Pseudo Log-Likelihood (PLL) gaps and moderate Adjusted Perplexity Index (APX) values reflected more balanced scoring between stereotype and anti-stereotype completions. The evaluation employed a Persian-adapted *StereoSet* dataset developed in this study, derived from the original English benchmark by localizing names, references, and sentence structures while preserving its four bias domains (Nadeem et al., 2021). This adaptation enabled culturally coherent bias measurement in Persian LLMs.

By contrast, LLaMA, trained primarily on English, showed directional bias in its native language, particularly in gender and religion-related prompts (Milios & BehnamGhader, 2022). When applied to Persian, it produced more

uniform and less expressive outputs, revealing limited sensitivity to linguistic and cultural nuances. This stems from its English-oriented tokenization and weak morphological grounding, which distort probability distributions and reduce bias sensitivity. While LLaMA achieves strong fluency in English, it does not generalize fairness across languages. PersianLLaMA, in contrast, displayed higher variance and more balanced scoring in Persian, suggesting that language-specific pretraining and tokenizer design enhance cultural and morphological sensitivity, thereby reducing, though not eliminating, bias in low-resource languages (Guo et al., 2024; Milios & BehnamGhader, 2022).

Addressing the second question, the results confirm that tokenization and language alignment strongly affect bias measurement and propagation. LLaMA's BPE tokenizer, optimized for Latin scripts, fragmented Persian text into incoherent subwords, leading to distorted PLL and Contextual Bias (CB) metrics (Milios & BehnamGhader, 2022). PersianLLaMA's Sentence-Piece tokenizer preserved morphological integrity, producing more stable probability distributions and interpretable bias metrics. These findings demonstrate that bias arises not only from training data imbalance but also from linguistic preprocessing and alignment choices.

Overall, this research contributes a reproducible cross-lingual framework for evaluating bias in Persian through PLL, CB, and APX metrics. The findings show that bias behaviors do not generalize symmetrically across languages; thus, linguistic and cultural alignment must be treated as primary design factors in multilingual LLMs. Future work should expand culturally adapted benchmarks with native human annotations, develop morphology-aware tokenization strategies, and integrate dynamic mitigation layers such as counterfactual data augmentation and adversarial debiasing for real-time fairness calibration (Guo et al., 2024).

By advancing these directions, future research can help build equitable, transparent, and culturally robust language technologies that reflect the linguistic and social diversity of global AI ecosystems.

Declaration of AI Assistance

In accordance with Stockholm University guidelines, I declare that AI-assisted tools (including ChatGPT) were used to support parts of the writing and linguistic refinement of this thesis. All scientific reasoning, experimental design, analyses, interpretations, and conclusions were conducted by the

author. AI tools were not used to generate research results, code, or experimental data, nor were they used to replace critical thinking or academic judgment. The author takes full responsibility for the accuracy and integrity of the content.

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